

Question Paper

Exam Date & Time: 27-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Deep Learning [IT_4442]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare the bias-variance tradeoff in machine learning. How do you balance the two when tuning a model to achieve better generalization? (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Analyze the differences between classification and regression in supervised learning. How do the techniques used in each differ in terms of model evaluation? (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Analyze the components of a deep feedforward network. How do the input, hidden, and output layers work together to learn from data? (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Apply the backpropagation algorithm to adjust weights in a neural network. How does the gradient of the loss function guide the training process? (5)
- Course Outcome: "CO5" / Cognitive Level: "Application"
- 3) Apply data augmentation techniques to improve the generalization of a deep learning model. How does augmentation help in preventing overfitting on limited datasets? (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Apply regularization techniques such as L1 and L2 to prevent overfitting. How do they affect the complexity of the model and its performance? (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Evaluate the importance of choosing an appropriate learning rate for neural network optimization. How does the learning rate schedule affect model training and convergence? (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Evaluate"
- B) Evaluate the role of activation functions in CNNs. How do they impact the learning ability of the network during training? (5)
- Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 5) Evaluate the impact of vanishing and exploding gradients in RNNs. How do these issues affect the network's ability to learn long-term patterns? (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- B) Evaluate the significance of the input, forget, and output gates in LSTMs. How do they contribute to the network's ability to remember long-term dependencies? (5)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4064:Deep Learning

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	15
4	4	CO 4	5
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_4064:Deep Learning

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	3	15(30%)
4	Analysis	3	15(30%)
5	Create	4	20(40%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)